

Lightweight AdaFace via AdaDistill

A Novel Approach to Knowledge Distillation
for Face Recognition

Based on the work of AdaFace (CVPR 2022)
and AdaDistill (ECCV 2024)

M11402161, M11402151, M11402152

The Journey So Far: Recapping the Mid-Term Work on AdaFace

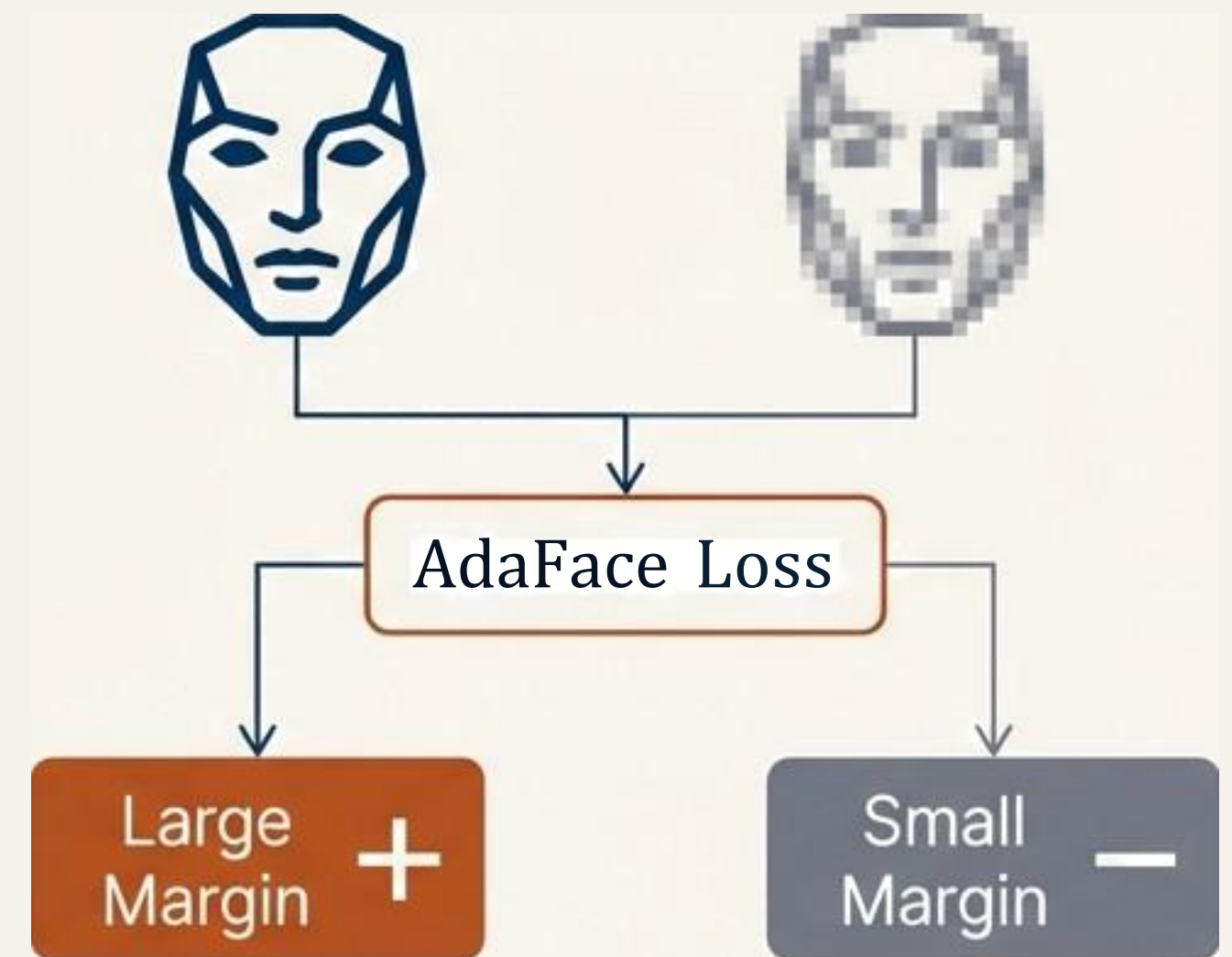
The Core Problem

Face recognition models struggle with varying image quality (e.g., blur, pose, lighting). Standard losses like ArcFace treat all images equally, leading to instability with low-quality samples.

The AdaFace Solution

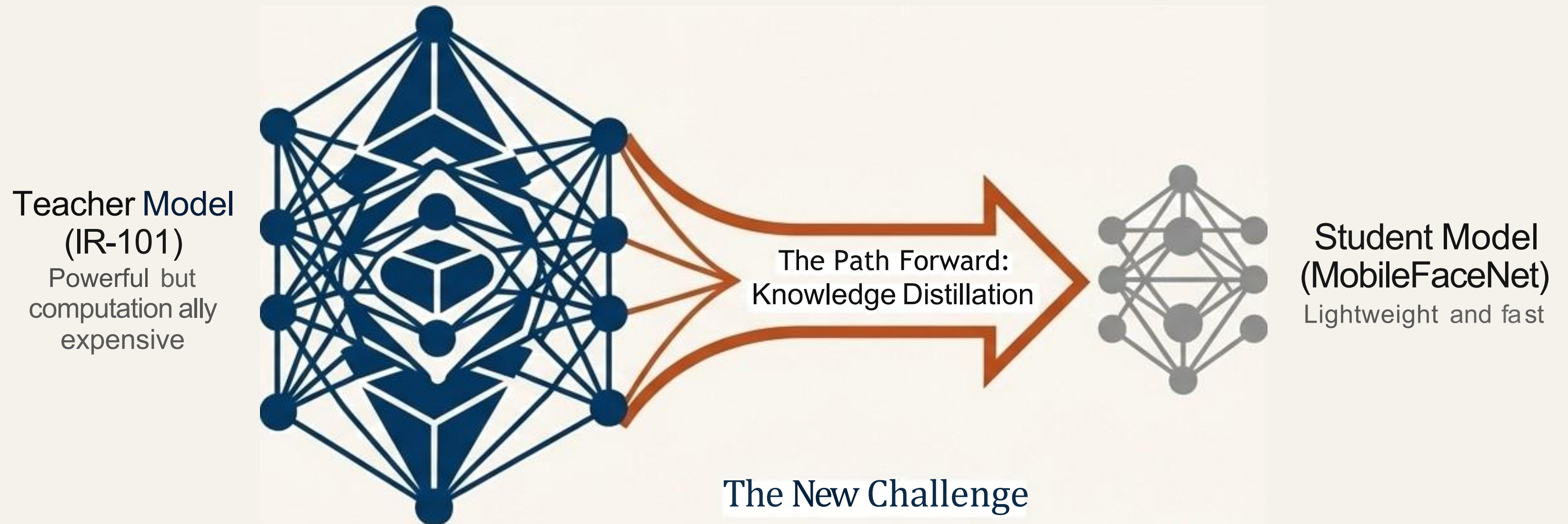
- Introduce an adaptive margin mechanism that accounts for sample quality.
- Uses the feature norm ($\|\mathbf{z}\|$) as a proxy for image quality.
- Dynamically scales the margin: larger for high-quality images, smaller for low-quality images.

AdaFace: Adaptive Difficulty



Key Takeaway: AdaFace stabilizes training and improves robustness by adapting the learning difficulty to the quality of the input face.

The Next Frontier: Making Quality-Aware Recognition Efficient



While powerful, models like IR-101 with AdaFace are computationally expensive. How can we transfer this nuanced, quality-aware knowledge to a much smaller, faster 'student' model like MobileFaceNet?

The Guiding Question: Can we design a distillation process that doesn't just mimic the teacher's output, but also teaches the student *how* to handle images of varying quality?

Midterm Result: IR-101+AdaFace

Dataset	AdaFace Paper (%)	Pretained (%) / Δ (Diff.)	Scratch (%) / Δ (Diff.)
LFW	99.83	99.80 / -0.03	99.8 / -0.03
CFP-FP	98.65	98.67 / +0.02	98.41 / -0.14
AgeDB-30	98.27	98.05 / -0.22	98.23 / -0.04
CPLFW	93.46	93.53 / +0.07	92.83 / -0.63
CALFW	96.08	96.12 / +0.04	96.05 / -0.03
TinyFace (Rank-1)	68.30	68.11 / -0.19	68.27 / -0.11
IJBB (TPR@FPR=0.01)	95.60	95.59 / -0.01	95.47 / -0.13
IJBC (TPR@FPR=0.01)	96.80	96.81 / +0.01	96.78 / -0.02

Our Philosophy: A More Sophisticated Student-Teacher Exam System

We frame our knowledge distillation process as an adaptive educational system.

The Key Players

The Student (f_s)

The lightweight MobileFaceNet model, providing its answer (embedding) for a face.

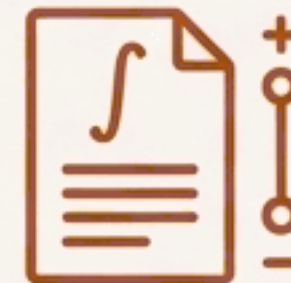


The Teacher (f_t)

The powerful IR—101/IR-50 model, providing the expert "demonstration answer."

The Standard Answer (w_y)

The class center for an identity.
Crucially, this is no longer fixed.



The Exam Paper (Loss Function)

The set of rules for grading the student, with a difficulty level (margin) that changes dynamically.

The Goal: The Teacher corrects the 'Standard Answer,' and the Student strives to match it on an 'Exam Paper' of varying difficulty.

The Teacher Corrects the Answer Key: Dynamic Class Centers

The Old Way (without a Teacher)

Class centers (w_y) are a “consensus” formed by all student samples over time.

This is unreliable in early training when the student is weak.

The AdaDistill Insight

A weak student shouldn't learn from an abstract consensus.

The “standard answer” itself should be guided by the teacher's expertise.

Mechanism: Exponential Moving Average (EMA)

$$w_{new} = \alpha' \cdot w_{old} + (1 - \alpha') \cdot f_t$$

Corrected Answer Key — w_{new} — Trust Factor — α' — Old Consensus — w_{old} — Teacher's Expert Opinion — f_t

The Teacher's Wisdom: Judging When to Intervene, and How Deeply

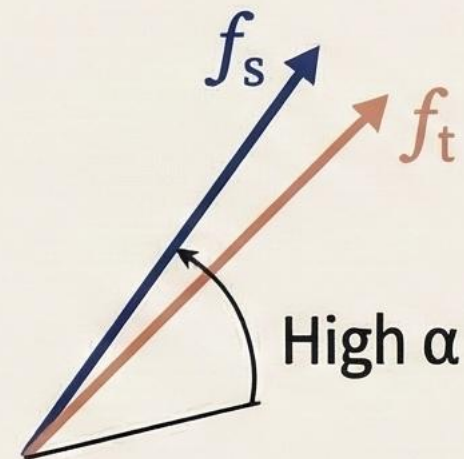
The size of α' is determined by a combination of two key indicators, assessing the status of the "student" and the "old standard answer".

Section 1: The Student — α

$$\alpha = \cos(f_s, f_t)$$

Measures how similar the 'student's answer' is to the 'teacher's demonstration'.

- High α : The student understands the question.
- Low α : The student does not understand, writes randomly.

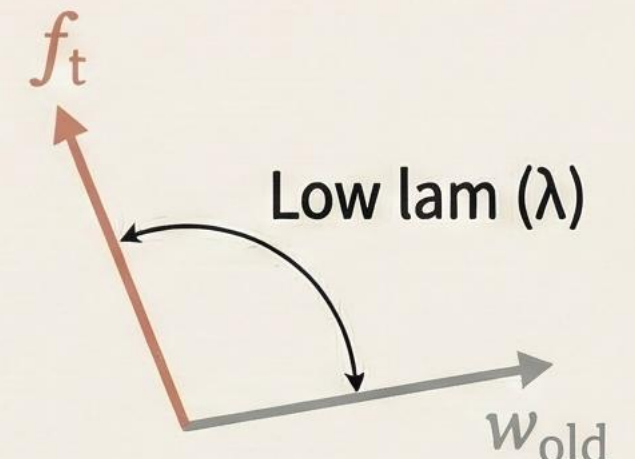


Section 2: The Standard Answer — lam (λ)

$$\text{lam} = \cos(w_{\text{old}}, f_t)$$

Measures how similar the 'old standard answer' is to the 'teacher's demonstration for the current example'.

- High lam: The teacher thinks the old answer is generally reliable.
- Low lam: The teacher thinks the old answer is itself wrong.



Final Decision Logic

$$\alpha' = \alpha \cdot \text{lam}$$

Only when "the student understands" AND "the old answer is reliable" will the system trust the old center. If either one has a problem, the teacher will take the lead in updating.

Three Teaching Scenarios: The Teacher's Decision Model

Scenario	α (Student Understanding)	$\lambda \alpha$ (Answer Quality)	α' (Combined Decision)	Action
Student Struggles	Low	High	Low	Trust the Teacher: The student is wrong, so the teacher personally corrects the standard answer.
Flawed Question	High	Low	Low	Trust the Teacher: The standard answer is problematic, so the teacher must rewrite it.
Great Success	High	High	High	Trust the Center: The student gets it, and the answer is right. Continue learning the class consensus.

$$w_{new} = \alpha' \cdot w_{old} + (1 - \alpha') \cdot f_t$$

$$\alpha' = \alpha \cdot \lambda$$

The Exam Adapts, Part 1: Difficulty Based on Question Clarity

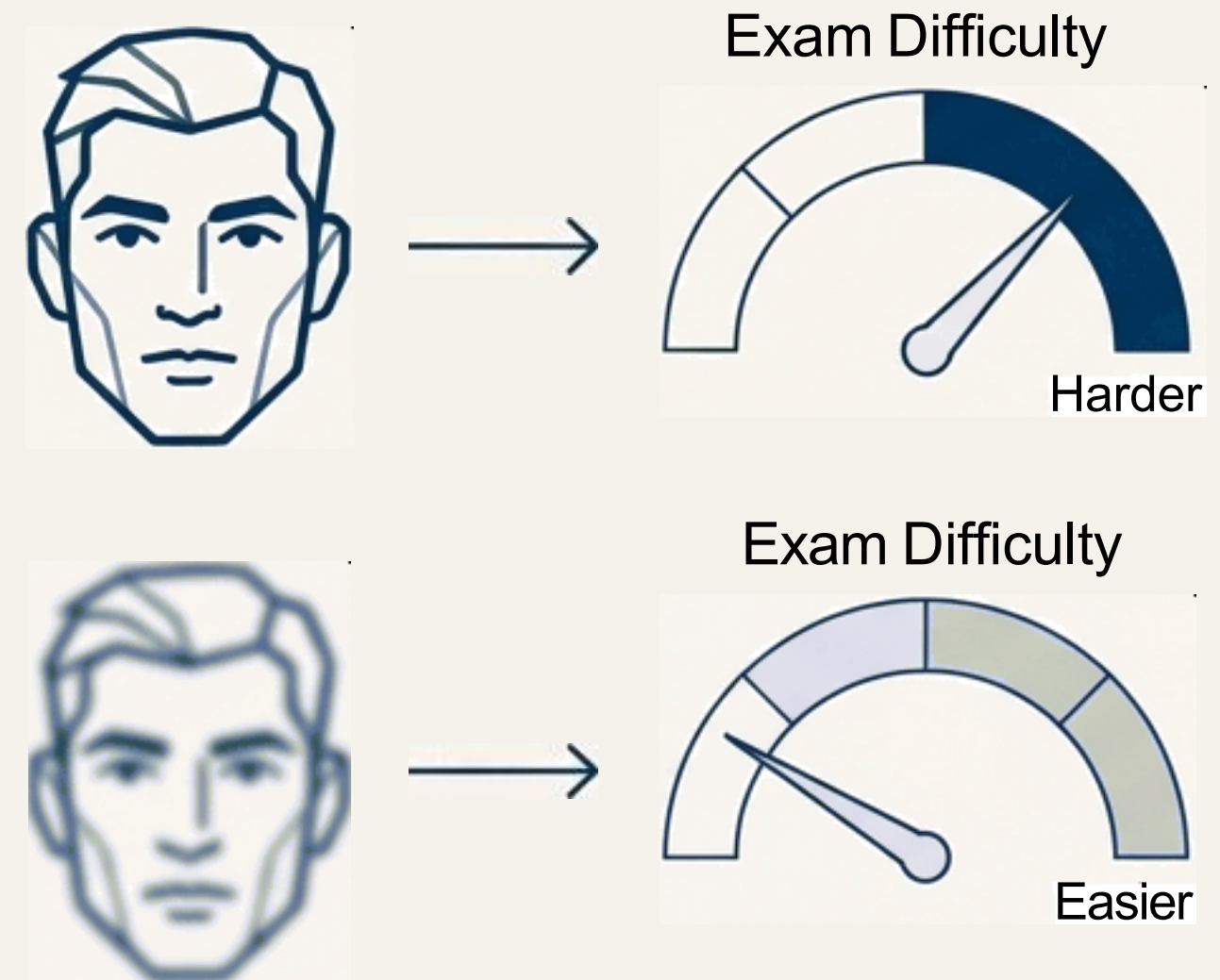
This mechanism is inherited directly from AdaFace and focuses on the input image itself.

Quality Proxy

The L2 norm of the feature embedding ($\|f\|$) serves as an indicator of image quality.

Dynamic Margin Logic

- **High-Quality Image** → High Feature Norm → **Larger Margin**.
The exam is made harder for clear, easy questions to push for better feature separation.
- **Low-Quality Image** → Low Feature Norm → **Smaller Margin**.
The exam is made easier for blurry, hard questions to prevent the model from overfitting to noise.



Impact: This only changes the *difficulty of the test* for the student (the logit), not the *standard answer* (the class center).

Comparison: ArcFace vs AdaFace

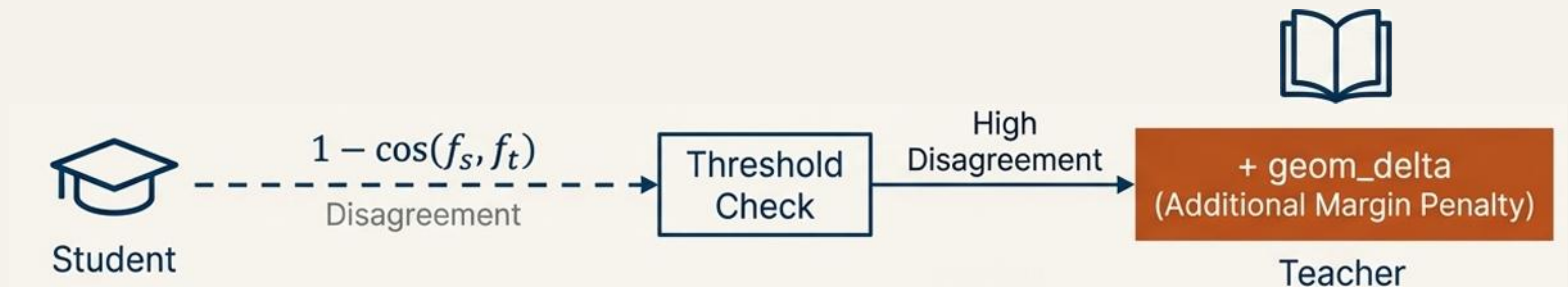
Method	LFW	CFPFP	AgeDB-30	CALFW	CPLFW	IJB-C (1e-4)	IJB-C (1e-5)
MFN + ArcFace	99.52	91.66	95.82	95.12	87.93	89.13	81.65
MFN + AdaFace	99.57	92.43	96.07	95.32	88.46	90.21	83.78

The Exam Adapts, Part 2: Getting Stricter When the Student Falls Behind

Image quality isn't the only factor. What if the student is simply learning incorrectly?

Introducing the Geometry-Aware Margin

We add a penalty based on the student-teacher disagreement.



Impact: This provides a second layer of adaptation. Even for a medium-quality image, if the student is far off track, the learning challenge is intensified.

The Exam Adapts, Part 2: Getting Stricter When the Student Falls Behind

Mechanism & Plain Language Logic

1. Calculate teacher-student difference: $1 - \cos(\mathbf{f}_s, \mathbf{f}_t)$. The larger this value, the greater the difference between student and teacher answers.
2. Convert this difference value (after some protective processing) into an additional margin penalty term **geom_delta**.
3. Add to the original margin:
 $\text{margin} \leftarrow \text{margin} + w \cdot \text{geom_delta}$

Plain Language Logic

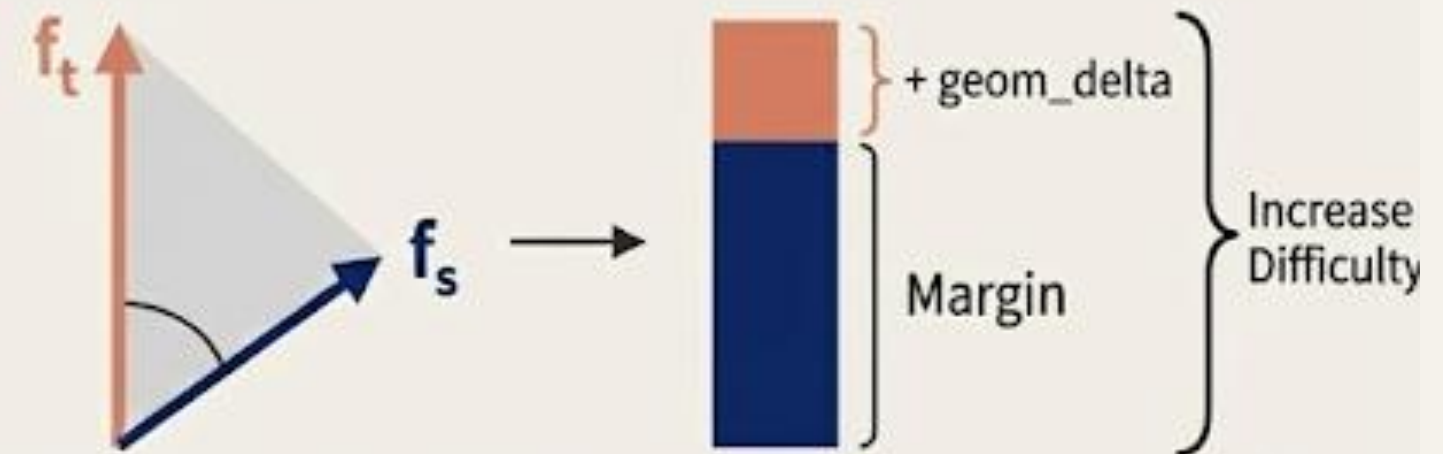
Originally: AdaFace only looked at “Is this face clear?”.

Now: If the student is too far from the teacher, we will say: “You’re learning too differently from the teacher, I’m going to increase the difficulty of this problem to force you back!”

Visual Explanation



Scenario 1: Small Student-Teacher Gap



Scenario 2: Large Student-Teacher Gap

The Flaw of the Static Approach: Hard-Coded Thresholds

The Problem:

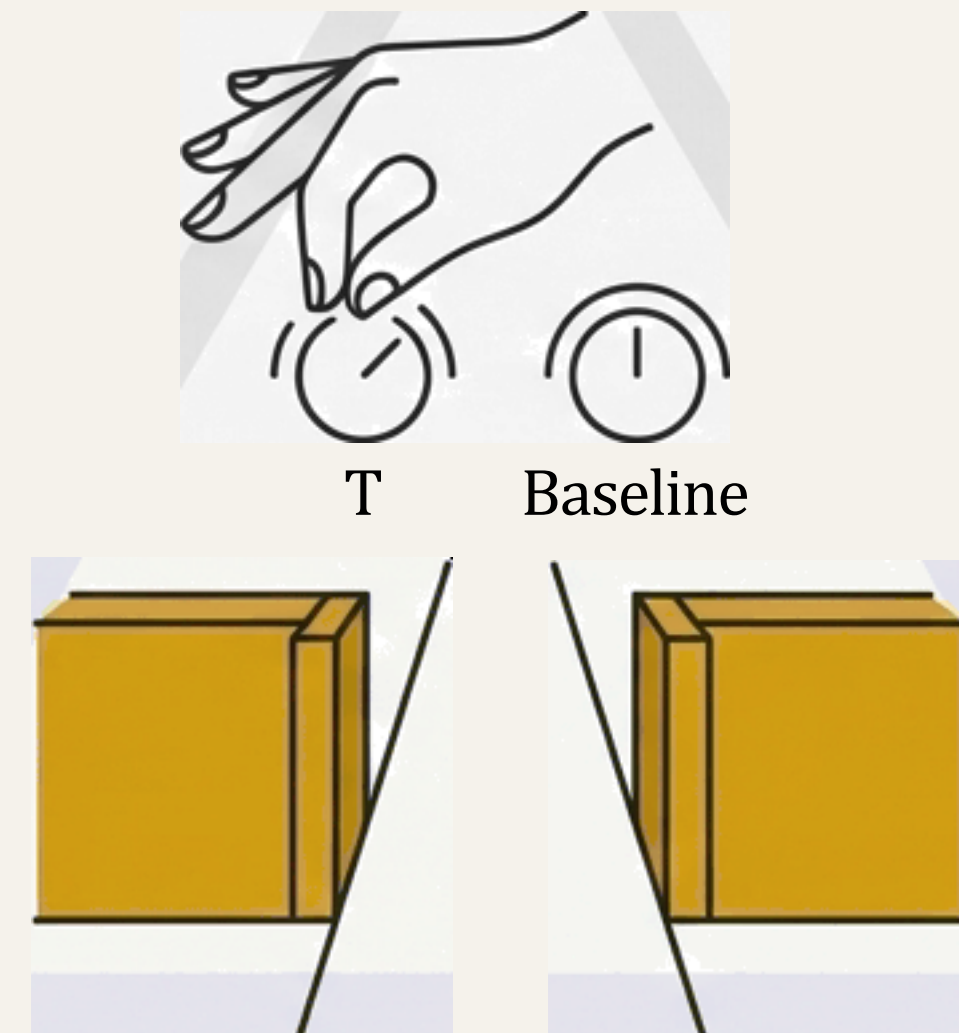
The static design required multiple protective mechanisms, which introduced manually-set hard thresholds.

The Thresholds:

1. Mask: 'If $\cos_st > i$ ' (e.g., 0.8), the student is 'close enough'. Do not apply a penalty.'
2. Baseline: 'Ignore minor deviations below a certain baseline, treating them as acceptable noise.'

The Consequence:

The model's performance becomes 'highly sensitive' to these hand-picked values, making it difficult to generalize across different datasets or teacher models.



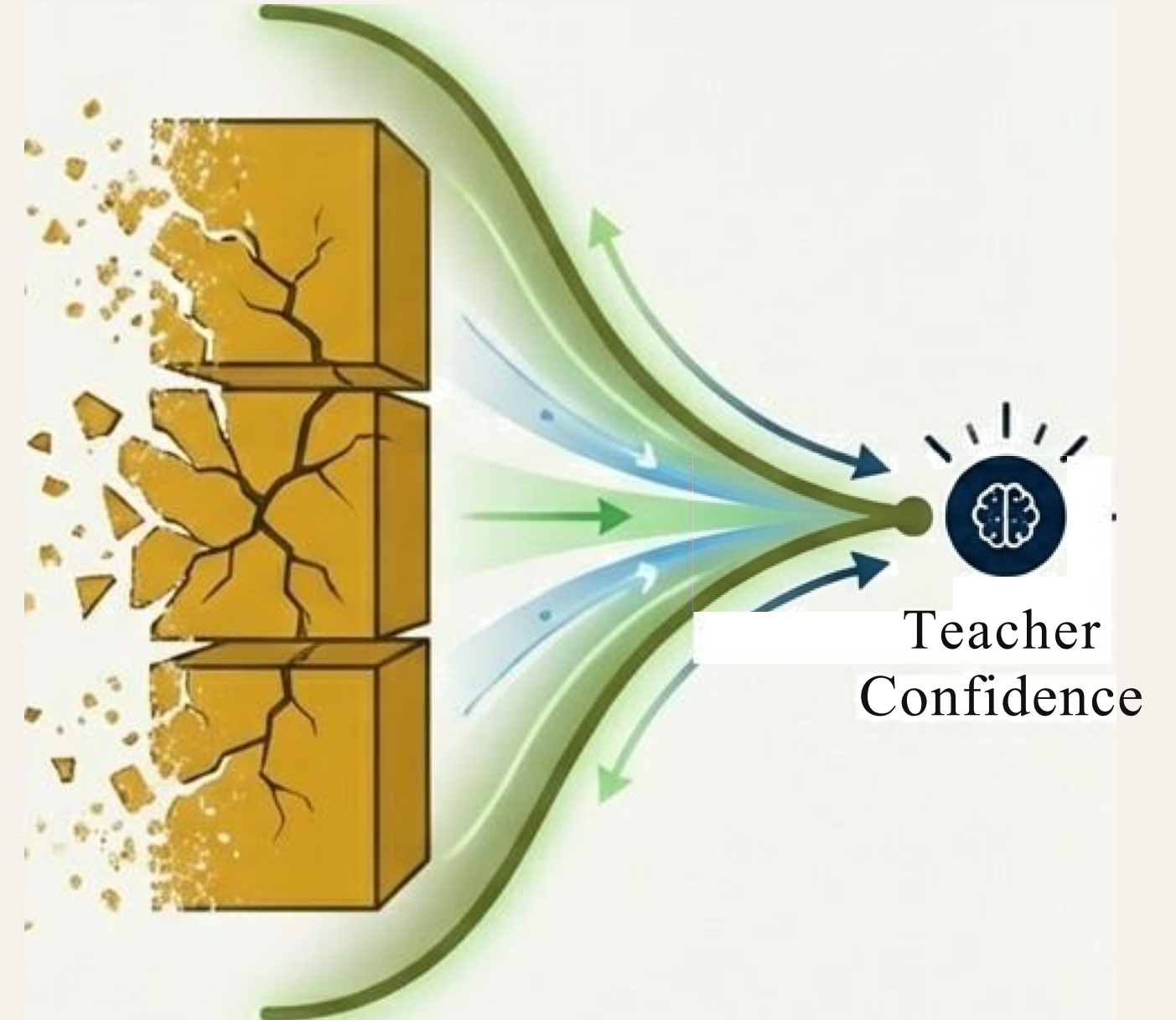
The Breakthrough: An Adaptive, Teacher-Guided Margin

The Core Insight:

Instead of relying on artificial hard thresholds, we can let the teacher model's own 'confidence' automatically determine the penalty conditions and strength.

The Result:

A new **adaptive geometry-aware margin** that eliminates the need for manual tuning and extra hyperparameters.



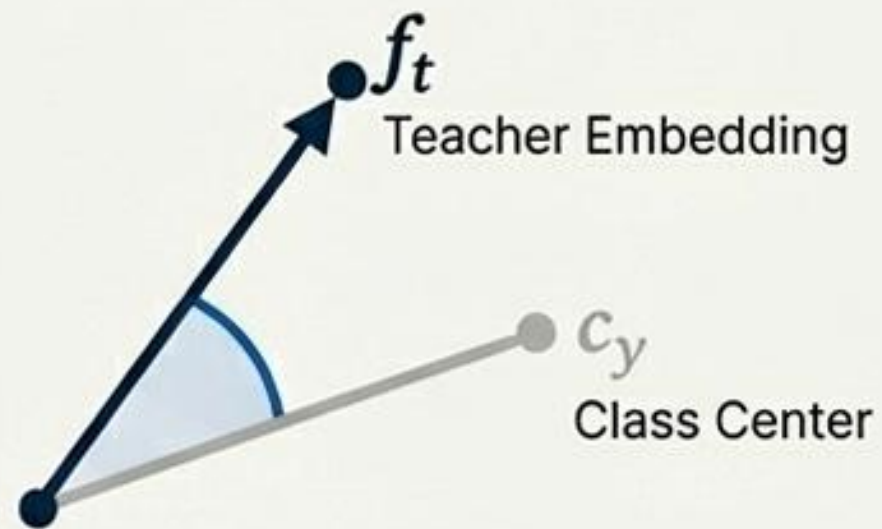
The Two Key Signals for an Adaptive System

Signal 1: Teacher Confidence (Conf_t)

How certain is the teacher about its own embedding?

We measure this as the cosine similarity between the teacher's embedding and its own correct class center.

$$\text{Conf}_t = \cos(f_t, c_y)$$

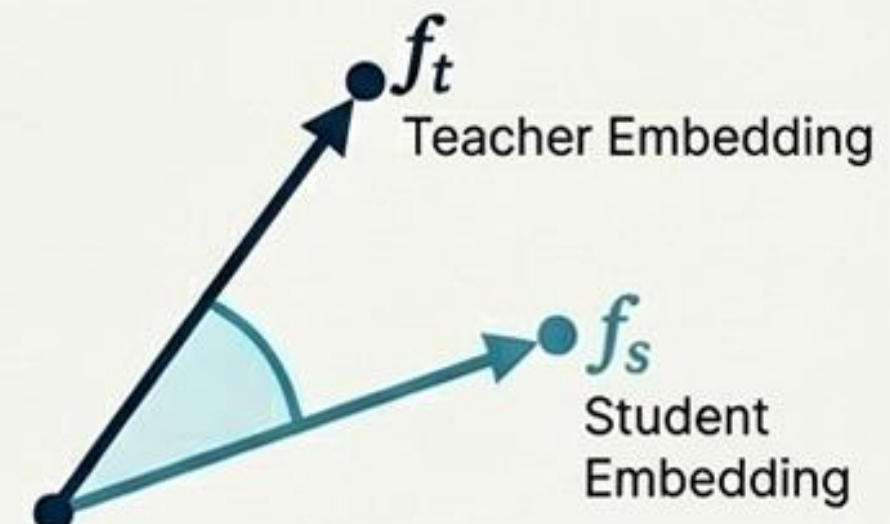


Signal 2: Student-Teacher Alignment (cos_{st})

How well is the student following the teacher's direction?

The cosine similarity between the student and teacher embeddings.

$$\text{cos}_{st} = \cos(f_s, f_t)$$



The Adaptive Penalty Formula

$$\text{Penalty} = \text{ReLU}(\text{Conf}_t - \text{cos}_{st}) \cdot \text{Conf}_t$$


Part 1: $\text{ReLU}(\text{Conf}_t - \text{cos}_{st})$

This term determines **WHEN** to apply a penalty. It acts as a dynamic, teacher-defined activation condition.

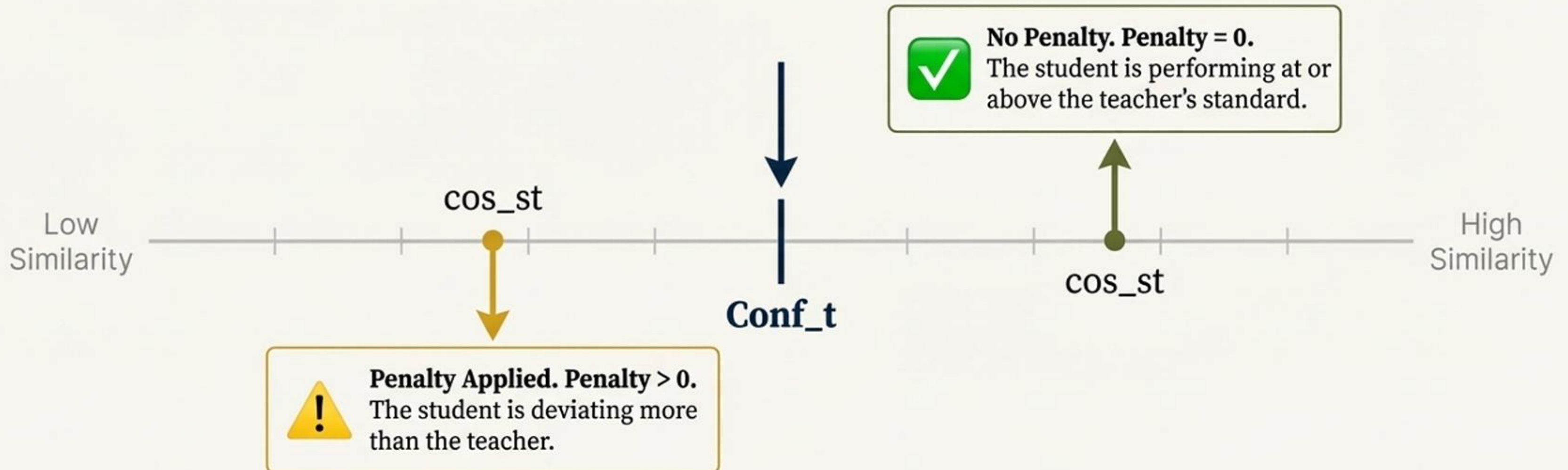
Part 2: $\cdot \text{Conf}_t$

This term determines **HOW MUCH** to penalize. It scales the penalty's magnitude based on the teacher's confidence.

Intuition 1: A Penalty is Applied Only When Necessary

A penalty is applied **only if** $\text{cos_st} < \text{Conf_t}$.

In Plain English: The student is penalized only if its embedding is “more misaligned” with the teacher than the teacher’s own embedding is with the class center.

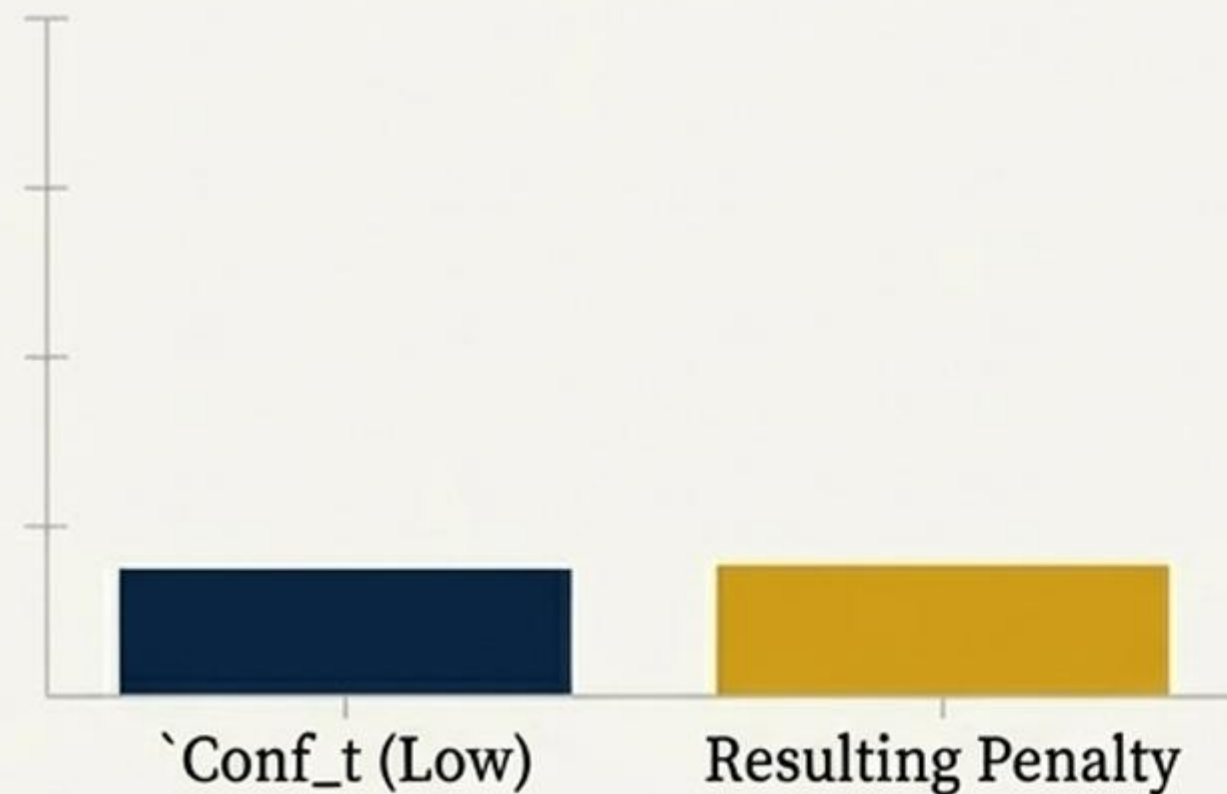


Intuition 2: Confident Teachers Are Stricter Teachers

The penalty's magnitude is scaled by the teacher's confidence (Conf_t).

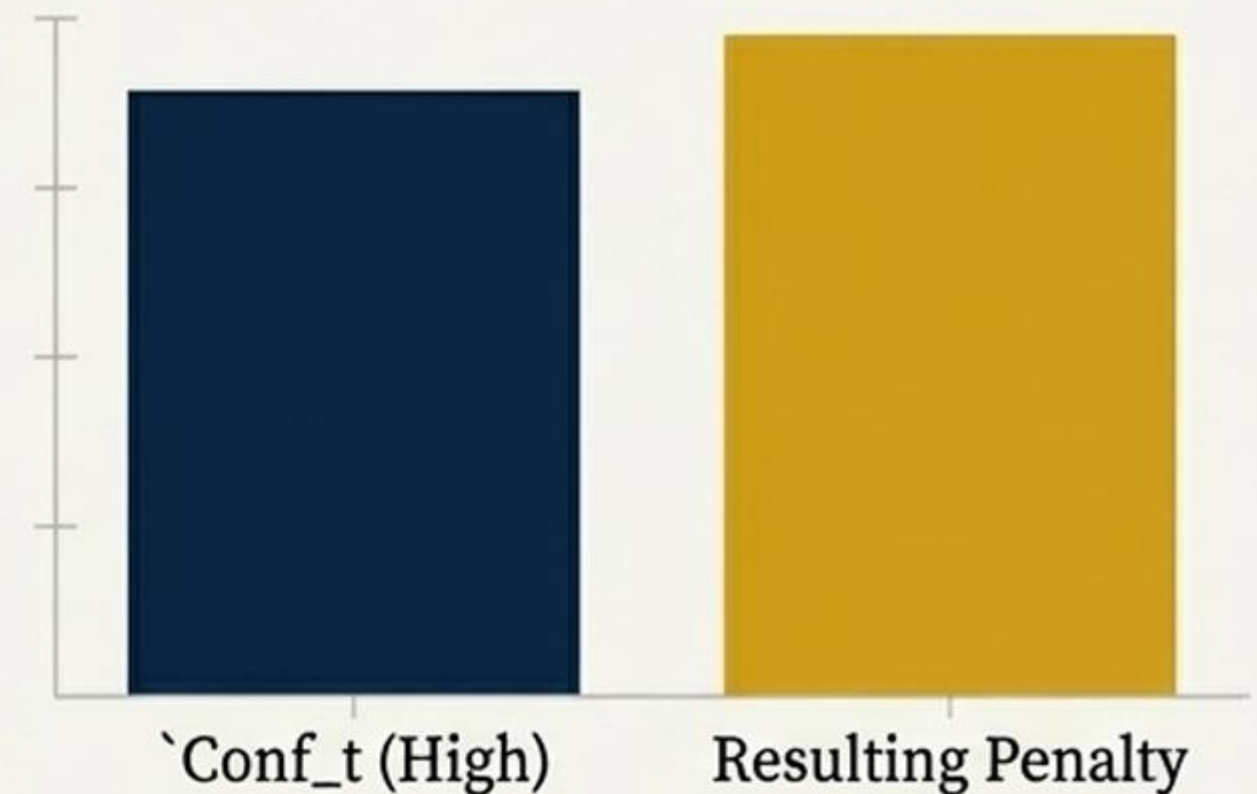
Scenario A: Teacher is Uncertain

Even if the student deviates significantly, the penalty is small. The teacher doesn't punish harshly for knowledge it isn't sure about.

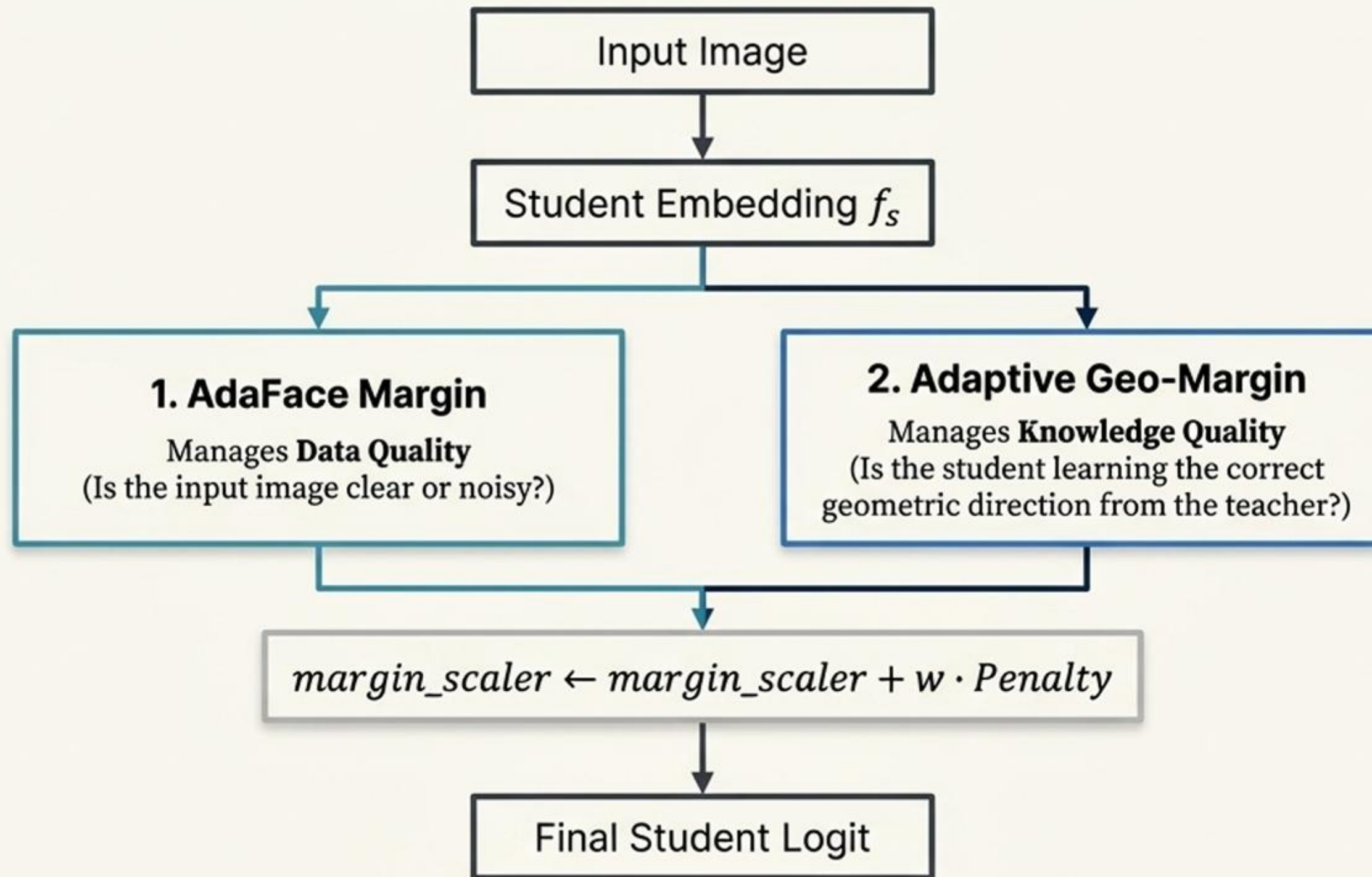


Scenario B: Teacher is Confident

Even a small student deviation is amplified into a significant penalty. The teacher strictly enforces its high-confidence knowledge.



A Unified System: Two Layers of Adaptive Control



Critical Clarification: Both mechanisms only act on the student's logit. The class center update remains exclusively controlled by AdaFace's original quality proxy.

From Static and Brittle to Adaptive and Robust

Feature	Static Geo-Margin (V1) ⚠️	Adaptive Geo-Margin (Final) ✅
Mechanism	Raw cosine difference	Teacher confidence as a guide
Thresholds	Multiple, hard-coded, manual	None. Zero extra hyperparameters.
Stability	Sensitive to tuning & data	Robust and self-regulating
Effort	High tuning cost	No tuning cost

The Impact: Superior and More Stable Performance

Experimental results confirm that the adaptive geometry-aware margin is significantly superior to the static version.

- ✓ **Improved Quality:** Effectively improves the quality of knowledge distillation.
- ✓ **Zero Tuning Cost:** Achieves better results without adding any new hyperparameters to tune.
- ✓ **Enhanced Tail Performance:** Shows notable improvement on difficult and tail samples.
- ✓ **Stable Training:** Eliminates a source of instability by removing hard thresholds.

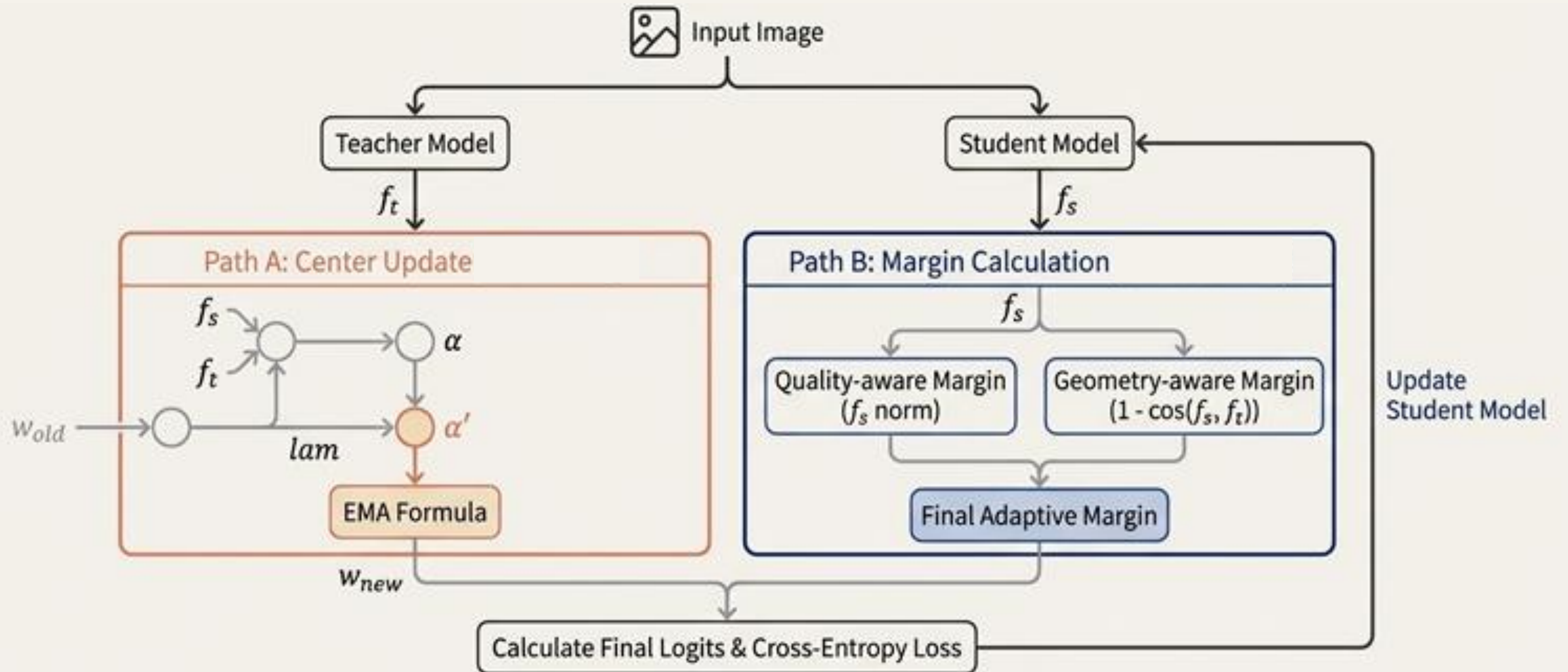
The Core Insight

By leveraging teacher confidence as a dynamic guide, we created an adaptive margin that intelligently corrects student model deviation without manual intervention.

This approach elegantly bridges the gap between managing data quality (via AdaFace) and ensuring knowledge alignment, leading to a more robust, stable, and effective knowledge distillation framework.



The Complete System: The AdaptiveAAdaFace Head



The Key Insight: The student is pushed towards a class center that is itself being corrected by the teacher. This creates an indirect but highly stable and robust learning process.

GitHub Contents

- **backbones/** – Backbone network architectures
- **config/** – Training configurations & hyperparameters
- **eval/** – Evaluation scripts
- **train/** – Core training pipeline (loss, dataloader, loops)
- **utils/** – Utility functions (logging, metrics, helpers)
- **img/** – Documentation images
- **output/** – Checkpoints & logs
- **requirements/** – Dependency files
- **run_AdaDistill.sh** – Main distillation launcher
- **run_AMLDistill.sh** – AML-based distillation launcher
- **run_standalone.sh** – Standalone training/testing script
- **README.md** – Project overview

The Proof: Our Smarter Teaching System Produces a Better Student

Experimental Setup:

Teacher: IR-50 Backbone

Student: MobileFaceNet

Training Data: MS1MV2 (-5.8M images)

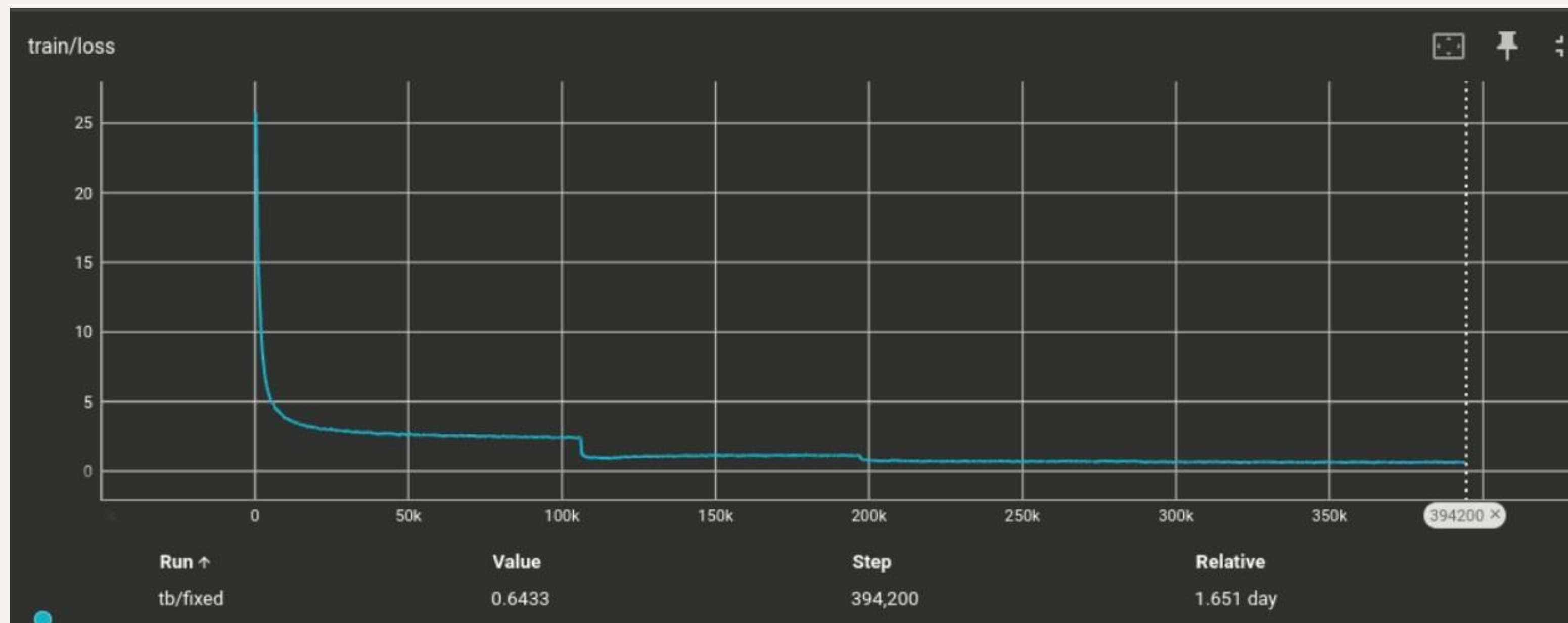
Performance Comparison: AdaDistill + AdaFace vs. the original MobileFaceNet

Dataset	MFN (%)	Distilled (%)	Diff (%)
LFW	99.57	99.567	-0.003
CFP-FP	92.43	95.329	+2.899
AgeDB-30	96.07	95.833	-0.237
CPLFW	88.46	90.150	+1.69
CALFW	95.32	95.133	-0.187
IJBC (TPR@FPR=1e-4)	90.21	93.823	+3.613
IJBC (TPR@FPR=1e-5)	83.78	90.004	+6.224

Observation

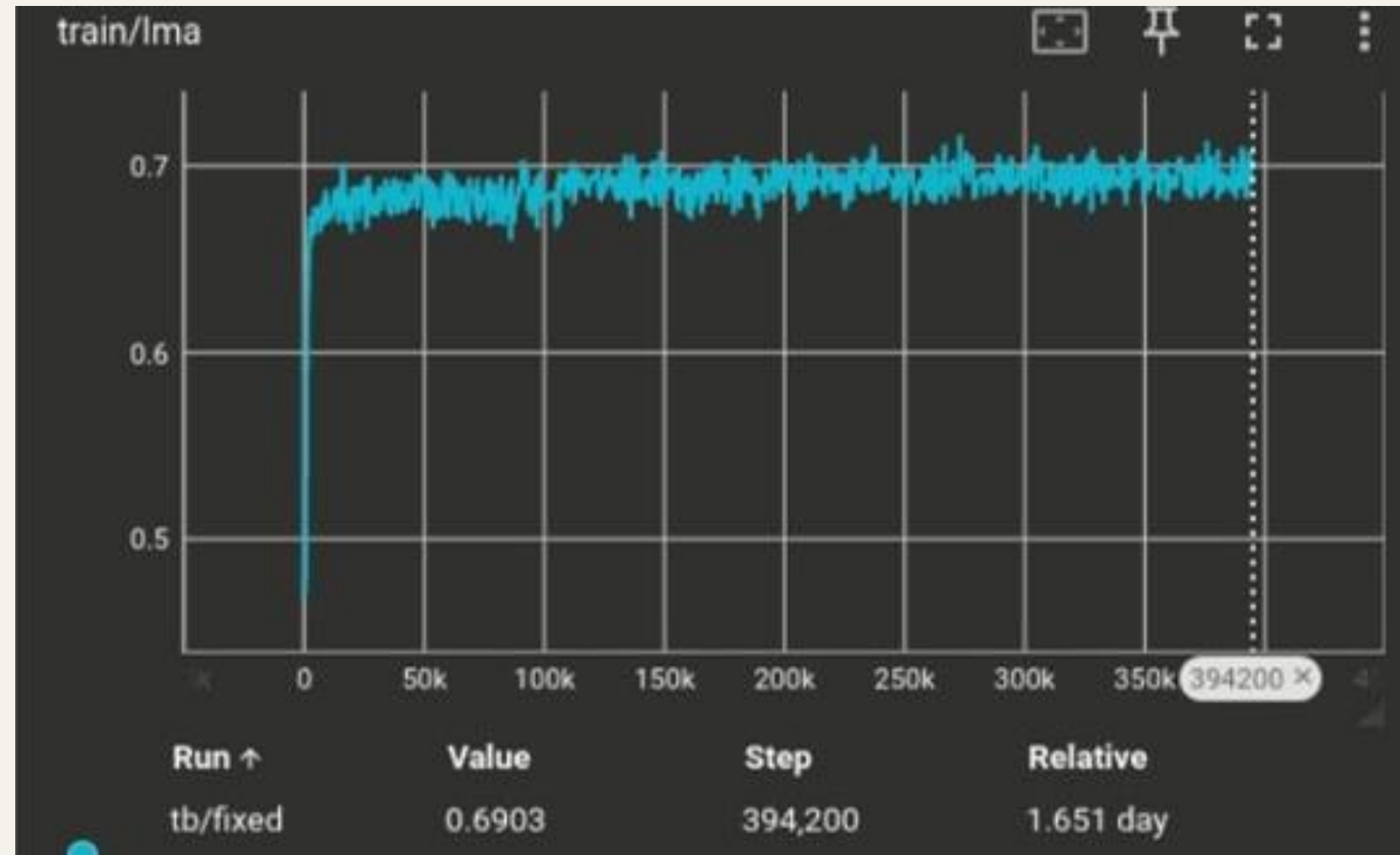
Significant improvements on challenging datasets like **TinyFace** and **CPLFW** demonstrate the method's effectiveness, especially for difficult, low-quality scenarios.

Experiment result

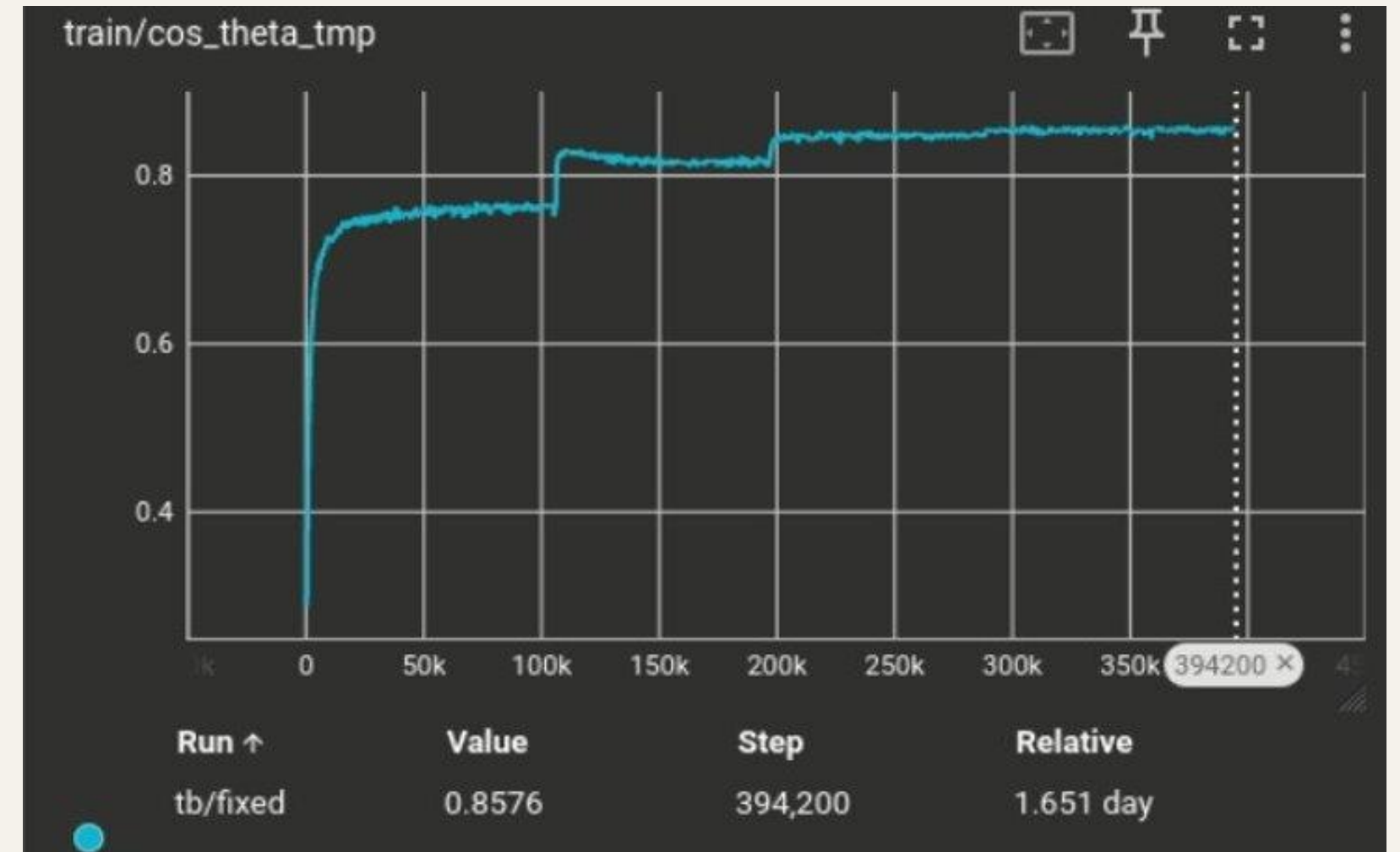


loss curve

Experiment result



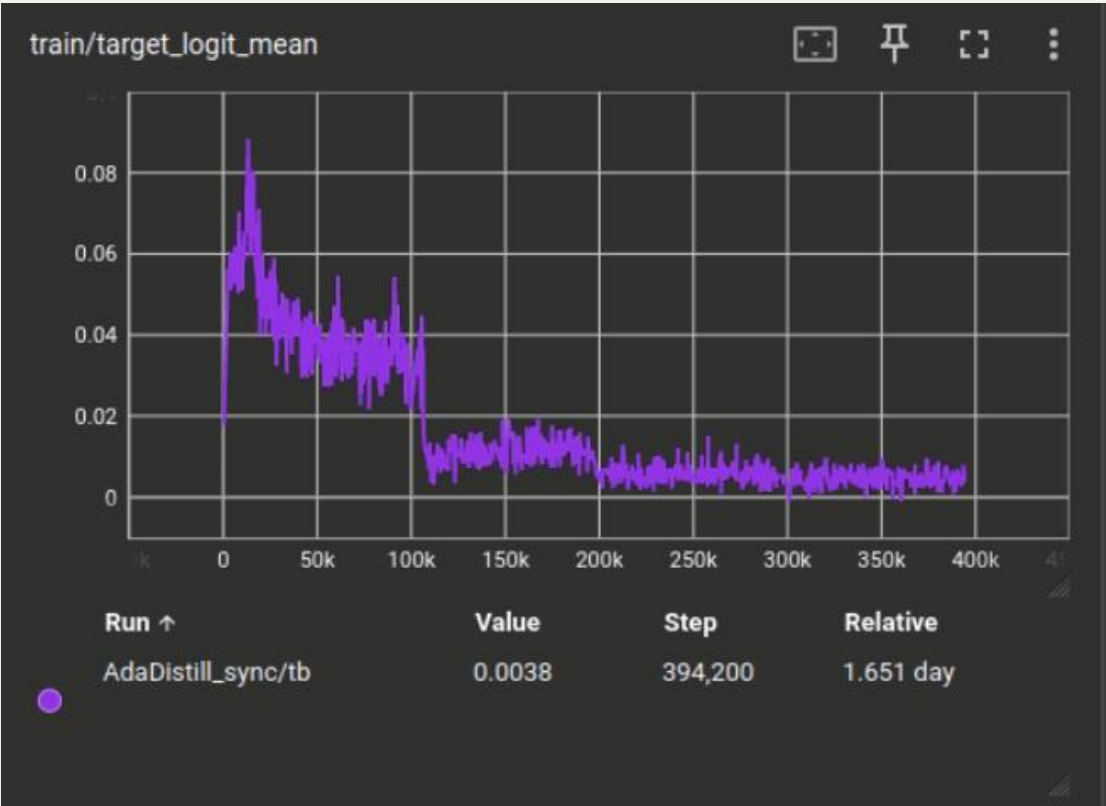
lma: lam mean average (α')



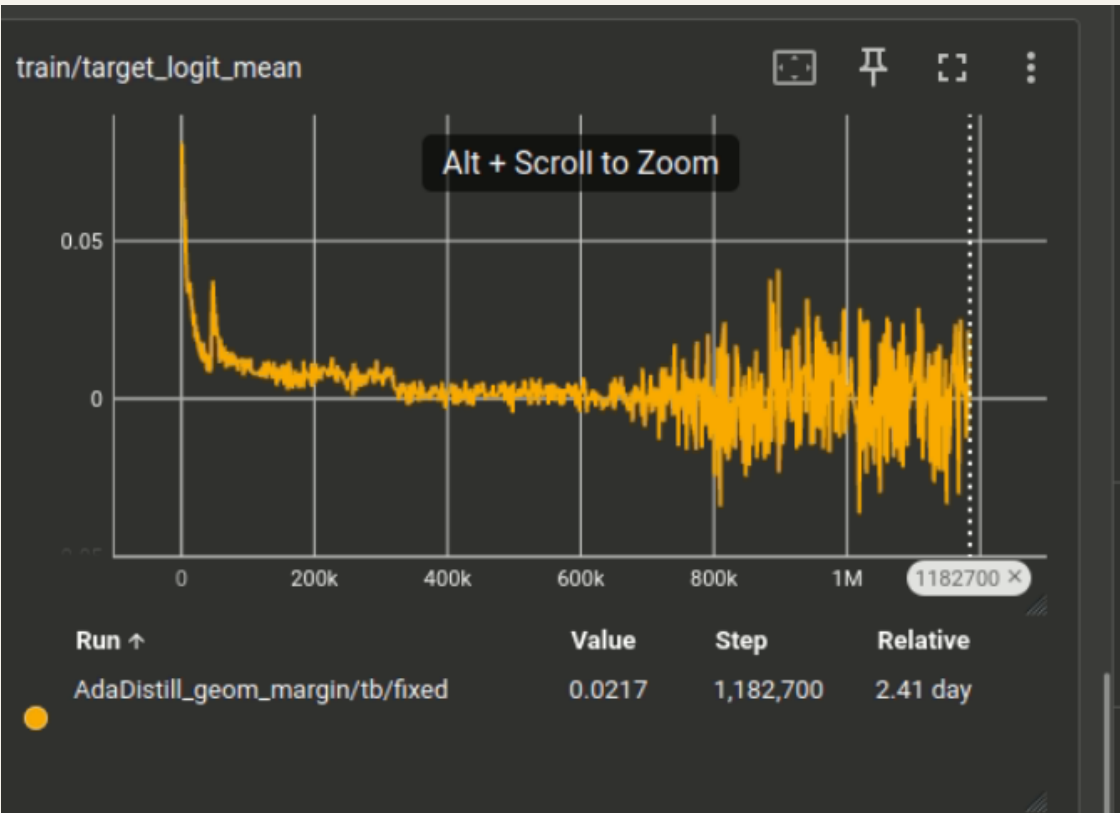
cos_theta_tmp (α)

Experiment result

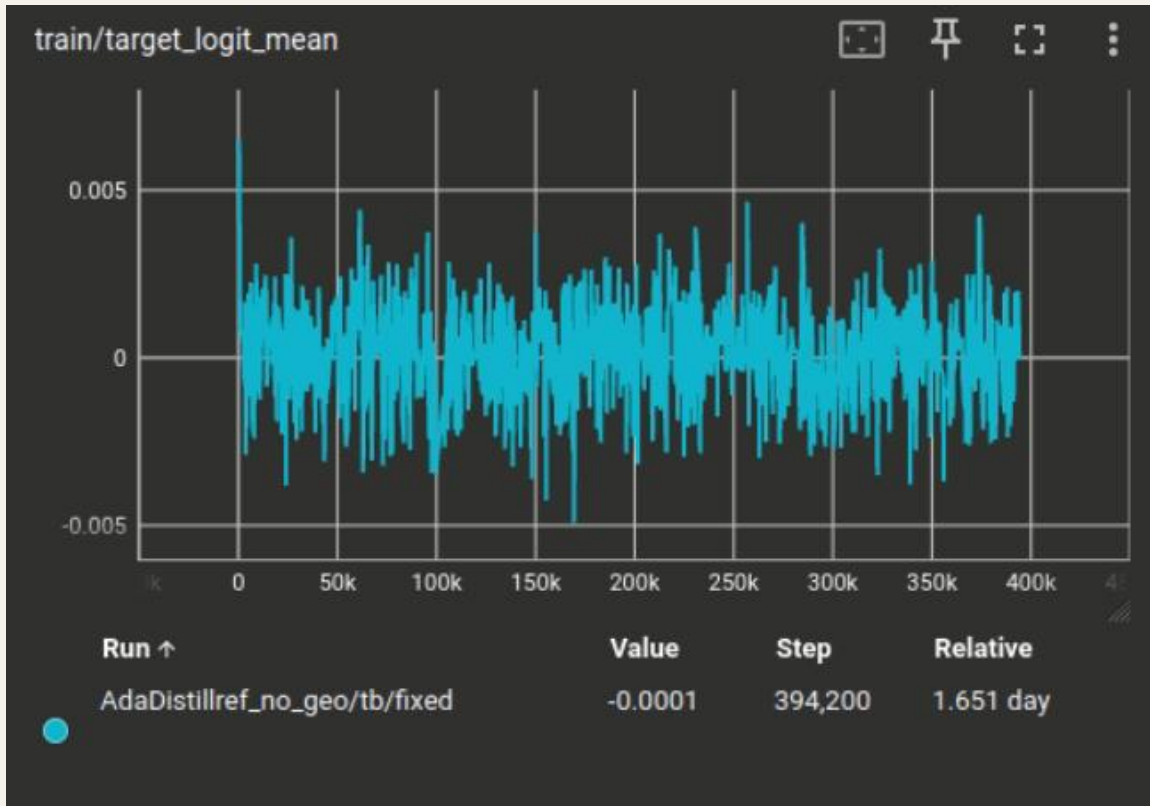
Logic Mean



adaptive geometry-aware



Static geometry-aware



No geometry-aware

Deeper Dive: Benchmarking Against a Strong Alternative

We compare our method (AdaD1sti1l + AdaFace) against the baseline student and 'AdaDistill' paired with the standard 'ArcFace' loss.

Performance on General Benchmarks

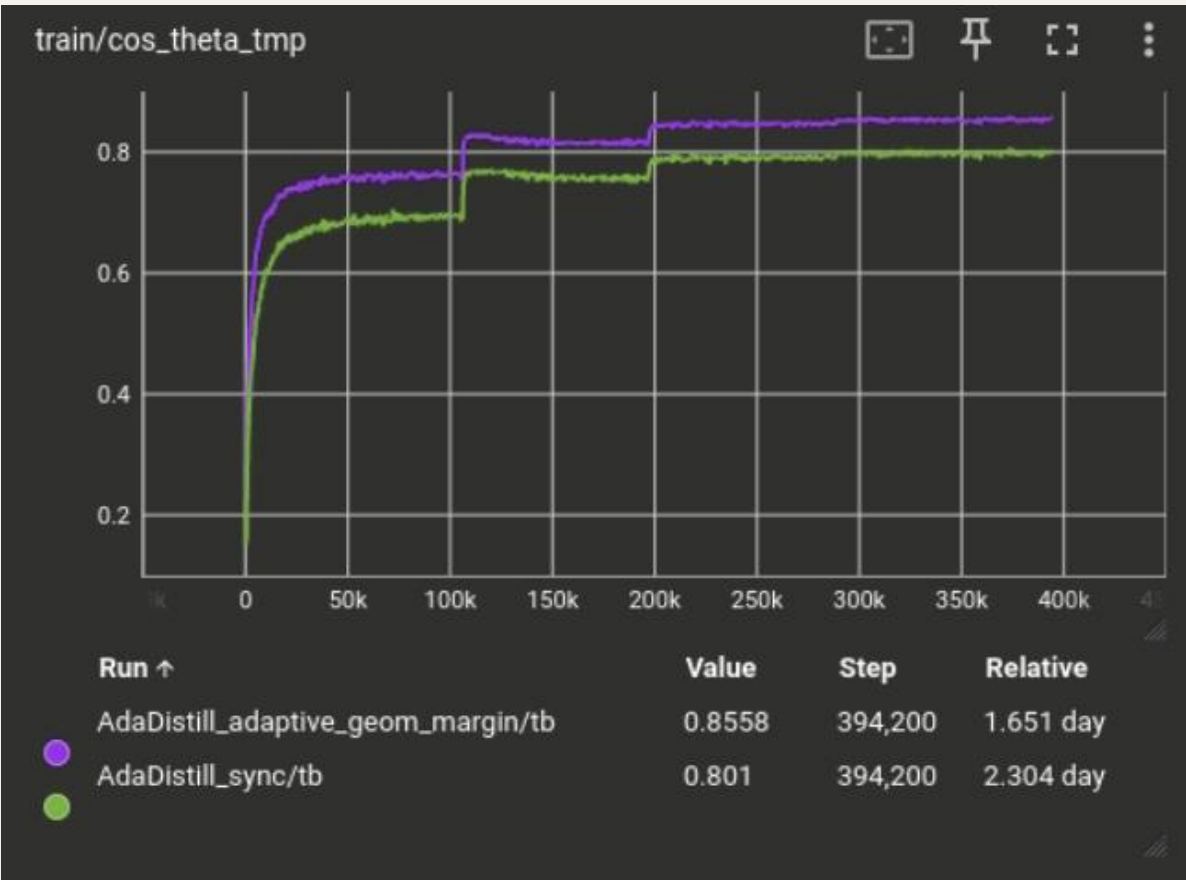
Method	LFW	CFP-FP	AgeDB-30	CA-LFW	CP-LFW	IJBC (1e-4)	IJBC (1e-5)	Avg
Baseline MFN (Student)	99.52	91.66	95.82	95.12	87.93	89.13	81.65	91.83
AdaDistill + ArcFace	99.47	95.14	96.68	95.50	89.50	93.26	89.31	94.12
AdaDistill + AdaFace	99.50	95.24	95.82	94.92	89.73	93.82	90.00	94.15
AdaDistill + AdaFace(Geo)	99.50	94.33	96.07	95.23	89.83	93.77	90.17	94.13
AdaDistill+AdaFace(adaptive_geo)	99.50	95.10	96.20	95.27	90.22	93.98	90.33	94.37

Performance on TinyFace (Low-Resolution)

Method	Tinyface R1	Tinyface R5
AdaDistill + ArcFace	57.21	63.12
AdaDistill + AdaFace	58.69	64.03
AdaDistill + AdaFace(Geo)	58.92	63.30
AdaDistill + AdaFace(adaptive_Geo)	59.31	64.43

Our integrated approach consistently outperforms both the standalone student and the standard distillation method, achieving the highest average performance.

Experiment result : Different Teacher Models (IR-101 vs. IR-50)



cos_theta_tmp (α)



lma: lam mean average (λ)

Teacher Influence: Is a Stronger Teacher Always Better for the Student?

Comparison of Distillation Results Using Different Teacher Models (IR-101 vs. IR-50)

	LFW	CFP-FP	AgeDB-30	CP-LFW	IJBC (1e-4)	IJBC (1e-5)	Tinyface R1
IR-101 (Teacher)	99.80	98.61	98.05	93.53	96.81	94.58	68.11
MFN (Student)	99.64	94.81	96.12	89.73	92.21	88.32	57.47
IR-50 (Teacher)	99.85	98.09	97.85	92.85	96.20	93.96	64.99
MFN (Student)	99.50	95.24	95.82	89.73	93.82	90.00	58.69

The student model cannot absorb the highly complex feature distribution learned by a very large teacher, causing weaker transfer. A mid-size teacher (ResNet50) produces better distillation results.

— Based on AdaDistill authors' explanation in the paper

Design Advantages: Why is This System More Stable and More Rational?

More Rational than ArcFace

We don't assume the class center is reliable from the start.

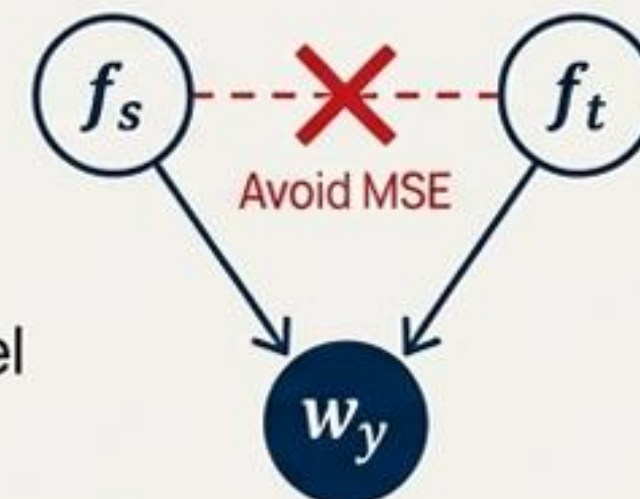
Instead, the center is gradually corrected by the teacher, providing a higher-quality learning target.



More Stable than “Direct Distillation”

We don't directly pull f_s towards f_t (e.g., via MSE loss). Instead, f_s learns from w_y , and f_t updates w_y .

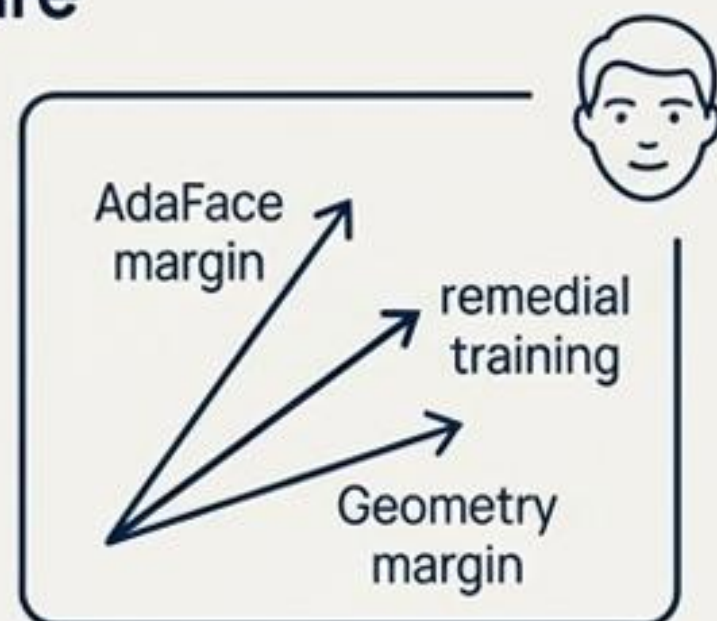
This preserves “class separability” and prevents the model from being distracted by noisy, single-instance teacher features.



Quality and Geometry Aware

For high-quality faces → use **AdaFace** margin for stronger learning.

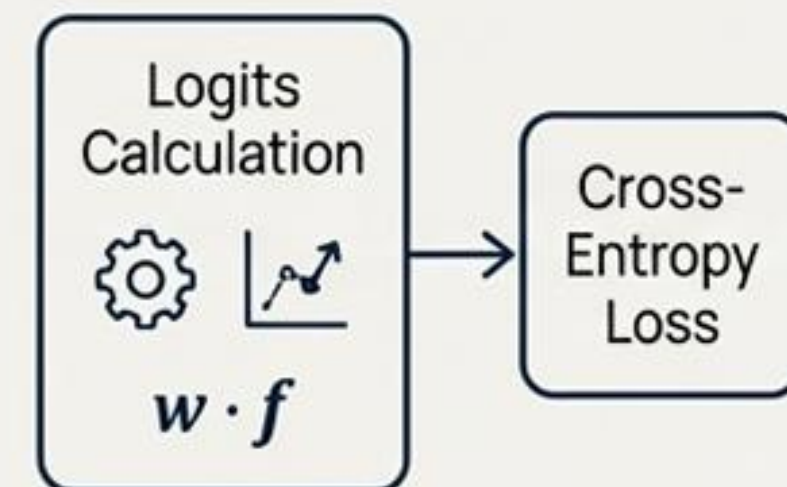
When the student has learned incorrectly → use Geometry margin for “remedial training.”



Simple to Implement

Ultimately, it's still a single cross-entropy loss.

All the complex dynamic adjustments are elegantly integrated into the logits calculation.



From Research to Real-World Deployment

Face Recognition Check-in System

Demo website: <https://checkin.rayyi-server.com/>

admin Password:123456

歡迎使用臉部辨識打卡系統

請選擇操作：

開始打卡

登入 / 查詢紀錄

註冊新帳號

使用者名稱

密碼 (至少 6 個字元)

確認密碼

臉部資料註冊 (請提供 3 個不同角度)

請確保光線充足、臉部清晰且位於畫面中央。

正在啟動攝影機...

擷取臉部照片 (0 / 3)

註冊

現有使用者

照片	ID	使用者名稱	管理員	操作
	1	admin	是	刪除 (自己)
	9	ray	否	刪除
	5	test1	否	刪除
	7	test2	否	刪除
	8	test3	否	刪除

返回管理員儀表板

應用程式設定

注意：更改某些設定（例如模型路徑、資料庫路徑、偵測/追蹤信心度）可能需要重新啟動伺服器才能完全生效。

設定項 (Key)	目前值 (Value)	類型 (Type)	新值 (New Value)
recognition_threshold	0.53	float	0.53
min_detection_confidence (需要重啟)	0.54	float	0.54
min_tracking_confidence (需要重啟)	0.5	float	0.5
onnx_model_filename (需要重啟)	adaptive_geo_backbone.onnx	str	adaptive_geo_backbone.onnx
blur_threshold	100	int	100
ratio_diff_threshold	0.25	float	0.25
recognition_mode	continuous	str	持續辨識 (Continuous) 鎖定辨識 (Lock-on)

儲存變更

返回儀表板

GPT Interaction Summary

Mid-Term Phase

- Structuring the presentation layout and data tables.
- Explaining configuration details and dataset issues (bcolz format, TinyFace path)
- Summarizing CSV results and preparing performance comparisons.
- Discussing the behavior of pretrained vs. from-scratch models.

Final Phase

- Implementing AdaFace within the AdaDistill repository architecture.
- Generating high-level explanations of the AdaDistill methodology.
- Assisting in the analysis and interpretation of final experimental results.

Resources and References

Code Repositories

- AdaFace: <https://github.com/mk-minchul/AdaFace>
- AdaDistill: <https://github.com/fdbtrs/AdaDistill>

Key Papers

- Kim et al. “AdaFace: Quality Adaptive Margin for Face Recognition.” (CVPR 2022)
- Boutros et al. “AdaDistill: Adaptive Knowledge Distillation for Deep Face Recognition.” (ECCV 2024)

Datasets Used

- Training: MS1MV2
- Evaluation: LFW, CFP-FP, AgeDB-30, CALFW, CPLFW, TinyFace, IJBB, IJBC